

# Fall 2022

**Problem 1 (Fa22).** The *mean oscillation* of a function  $f$  on an interval  $(a, b)$  is defined to be the average value of  $|f(x) - \bar{f}|$  on  $(a, b)$ , where  $\bar{f}$  is the average value of  $f$  on  $(a, b)$ . Calculate the mean oscillation of the function  $f = \log x$  on the interval  $(0, b)$ .

**Problem 2 (Fa22).** Let  $I$  be an open interval, and let  $f: I \rightarrow \mathbb{R}$  be a differentiable function. Assume that  $f'(x) \geq 0$  for all  $x \in I$ , and that all zeros of  $f'$  are isolated. Show that  $f$  is strictly increasing on  $I$ .

**Problem 3 (Fa22).** Suppose that  $f: \mathbb{R} \rightarrow \mathbb{R}$  is a continuous function from the reals to the reals, and let  $X \subseteq \mathbb{R}$  be a subset of the real numbers. For each of the following statements either prove it is always true or give an example to show it can be false.

1. If  $X$  is open then  $f(X)$  is open.
2. If  $X$  is closed then  $f(X)$  is closed.
3. If  $X$  is open then  $f^{-1}(X)$  is open.
4. If  $X$  is closed then  $f^{-1}(X)$  is closed.

**Problem 4 (Fa81, Fa17, Fa22).** 1. How many zeros does the function  $f(z) = 3z^{100} - e^z$  have inside the unit circle (counting multiplicities)?

2. Are the zeros distinct?

**Problem 5 (Fa22).** Let  $f$  be an analytic function on the real line, and define  $g$  by  $g(s) = \int_0^1 f(t)t^s dt$  for  $s > 0$ . Prove that  $g$  can be extended to a meromorphic function on the whole complex plane. Where would you expect to find the poles of  $g$ ?

**Problem 6 (Fa81, Sp13, Fa22).** Prove the following three statements about real  $n \times n$  matrices.

1. If  $A$  is an orthogonal matrix whose eigenvalues are all different from  $-1$ , then  $I + A$  is nonsingular and  $S = (I - A)(I + A)^{-1}$  is skew-symmetric.
2. If  $S$  is a skew-symmetric matrix, then  $A = (I - S)(I + S)^{-1}$  is an orthogonal matrix with no eigenvalue equal to  $-1$ .
3. The correspondence  $A \leftrightarrow S$ , called the Cayley transform, from Parts 1 and 2 is one-to-one.

**Problem 7 (Fa22).** Let  $A \in M_{n \times n}(\mathbb{F})$ , where  $\mathbb{F}$  is a field. Let  $T : M_{n \times n}(\mathbb{F}) \rightarrow M_{n \times n}(\mathbb{F})$  take  $X$  to  $AX$ . Compute the determinant and trace of  $T$  in terms of the determinant and trace of  $A$ .

**Problem 8 (Fa22).** Prove that any group of order 18 is a *semidirect product* of a normal subgroup of order 9 by a group of order 2, and use this to classify these groups.

**Problem 9 (Su80, Sp12, Sp21, Sp22, Fa22).** Let  $G$  be a finite group and  $H$  be a subgroup.

1. Show that the number of subgroups of  $G$  conjugate to  $H$  divides the index of  $H$ .
2. Show that if  $G = \bigcup_{g \in G} gHg^{-1}$  then  $G = H$ .

**Problem 10 (Fa22).** Find continuous bounded functions  $f_n$  on the reals (for  $n \geq 0$ ) such that all integrals and sums below are well defined and finite, but

$$\int_0^\infty \sum_{n=0}^\infty f_n(x) \, dx \neq \sum_{n=0}^\infty \int_0^\infty f_n(x) \, dx$$

**Problem 11 (Fa22).** Show that it is possible to find an uncountable set of subsets of the positive integers that is *totally ordered by inclusion* – given two of these subsets  $M, N$  either  $M \subseteq N$  or  $N \subseteq M$ .

**Problem 12 (Fa22).** Suppose that the functions  $g, f_1, f_2, \dots$  are smooth real valued functions on  $\mathbb{R}$  such that they all vanish at 0 and  $\lim_{n \rightarrow \infty} f'_n(x) = g'(x)$  for all  $x$ . Either prove or give a counterexample to the claim that  $\lim_{n \rightarrow \infty} f_n(x) = g(x)$  for all  $x$ .

**Problem 13 (Fa90, Fa22).** Let the function  $f$  be analytic in the entire complex plane, and suppose that  $f(z)/z \rightarrow 0$  as  $|z| \rightarrow \infty$ . Prove that  $f$  is constant.

**Problem 14 (Sp80, Fa22).** Let  $a > 0$  be a constant  $\neq 2$ . Let  $C_a$  denote the positively oriented circle of radius  $a$  centered at the origin. Evaluate

$$\int_{C_a} \frac{z^2 + e^z}{z^2(z-2)} \, dz$$

**Problem 15 (Fa22).** Let  $A$  be the  $n \times n$  complex matrix whose diagonal entries are all  $a$  and whose nondiagonal entries are all  $b$ . Find the eigenvalues of  $A$  and their multiplicities.

**Problem 16 (Su77, Fa22).** Let  $A$  be an  $r \times r$  matrix of real numbers. Prove that the infinite sum

$$e^A = I + A + \frac{A^2}{2} + \cdots + \frac{A^n}{n!} + \cdots$$

of matrices converges, i.e., for each  $i, j$ , the sum of  $(i, j)^{th}$  entries converges, and hence that  $e^A$  is a well defined matrix.

**Problem 17 (Fa90, Fa22).** Prove that  $\sqrt{2} + \sqrt[3]{3}$  is irrational.

**Problem 18 (Fa97, Fa22).** A finite abelian group  $G$  has the property that for each positive integer  $n$  the set  $\{x \in G \mid x^n = 1\}$  has at most  $n$  elements. Prove that  $G$  is cyclic, and deduce that every finite field has cyclic multiplicative group.